

VLE and flash calculations

After midterm · Session 1 of 6 · 180 minutes

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The question for today

How does pressure change the phase split at fixed temperature?

Write your prediction before opening the lab. By the end, support an answer with one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion and exit question	30

The 70-minute investigation is student work with instructor support.

Ideal-mixture boundary equations

$$P_{bubble} = \sum_i x_i P_i^{sat}$$

$$\frac{1}{P_{dew}} = \sum_i \frac{y_i}{P_i^{sat}}$$

Both equations are for fixed T, ideal liquid and ideal vapor. Using the same numerical composition in each equation describes different incipient-phase states.

TP flash includes a material balance

$$z_i = (1 - \beta)x_i + \beta y_i, \quad y_i = K_i x_i$$

$$\sum_i \frac{z_i(K_i - 1)}{1 + \beta(K_i - 1)} = 0$$

Here β is vapor mole fraction. In a nonideal liquid, K depends on the unknown x , so composition and phase fraction must be solved together.

A reproducible ideal baseline

Lab 01 synthetic binary at 350 K: $P_1^{sat} = 150$ kPa and $P_2^{sat} = 60$ kPa. Choose ideal liquid, $z_1 = 0.5$, $P = 95$ kPa.

QUANTITY	EXPECTED RESULT
Bubble P at $x_1=0.5$	105 kPa
Dew P at $y_1=0.5$	85.7143 kPa
Flash x_1, y_1	0.388889, 0.614035
Vapor fraction β	0.493506

The values are synthetic definitions, not real-fluid measurements.

An independent check

Independently close both component balances and confirm that all phase compositions sum to one.
Explain the phase state using bubble and dew pressures.

One demonstration • 20 minutes

Open the core lab

Use the synthetic ideal mixture at $T=350\text{ K}$, $P=95\text{ kPa}$ and $z_1=0.5$. Save the baseline, then change only P to 110 kPa . Predict the phase before calculating.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 01, save one ideal two-phase baseline and one changed pressure at the same T and z . Predict the change, report phase amounts and check one component balance by hand.

Discussion and exit question • 30 minutes

Use 20 minutes to compare results and discuss unexpected behavior.

How does pressure change the phase split at fixed temperature?

Use the final 10 minutes to explain your answer individually and name one assumption that limits it.

Required practice • 45–60 minutes

In Lab 01, save one ideal two-phase baseline and one changed pressure at the same T and z . Predict the change, report phase amounts and check one component balance by hand.

Assessment 1 and independent checkpoint

Retain one study JSON, one comparison and a short explanation. Continue from your classroom case. The hand checkpoint is part of this time budget.

Optional study

Choose one extension: compare liquid activity models, examine zero versus nonzero virial coefficients in Lab 09, or add the energy balance in Lab 07.

[Session appendix](#) · [Module reference](#)

[Learning path](#) · [Offline package](#)

Optional work is additional. Complete the required investigation first.